

Francisco Rau

AI Engineer · Agents, Prompting & Integrations

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Summary

Builder with an MSc in Data Science & Society and a BSc in Economics (Tilburg University) who ships AI-native software end to end. I solo-built Citania, a production multi-tenant SaaS: ~80,000 lines of TypeScript on PostgreSQL, 294 automated tests, contract-first OpenAPI/Zod, and live integrations with OpenAI, Stripe, Google Calendar, Microsoft Graph and WhatsApp. At Netvice I introduced agentic AI development as a delivery method and used it to build the company's real-time operations tooling. Prompt design, iteration and checking whether output actually holds up are daily work for me. Everything below is live at franciscorau.com.

Skills

AI: LLM application development, prompt engineering & iteration, output evaluation, agentic workflows (Claude Code, Replit), OpenAI API integration

Engineering: TypeScript/Node, React, PostgreSQL, REST APIs & webhooks, contract-first OpenAPI/Zod, Git, Vitest (294 tests), Python (analysis & modelling), SQL

Delivery: fast shipping under full ownership, stakeholder translation, process analysis, forecasting & statistics

Experience

Data & Process Analyst — Netvice

May 2025 – Present · Tilburg, Netherlands

- Introduced agentic AI development (Replit, Claude Code) as a delivery method — an early bet that let one analyst design, build, and ship production tooling end to end.
- Fully self-directed: each of the systems below began as my own proposal — need identified, pitched, built, and shipped at pace.
- Real-time operations layer: live dashboards wired to telephony webhooks (agent line state, idle detection, coverage views), plus a Daily Agent Planner, agent day reports, and a finance board (pilot).
- Built everything configurable rather than hard-coded, so the tooling keeps working as campaigns and staff change.
- Grew from data intern to Data & Process Analyst, working with increasingly senior stakeholders — ultimately on strategic projects directly with the company's founder.
- Established operational BI as a new, self-owned workstream — a first in the company's ten-year history — with full ownership from roadmap to daily use on the call-center floor.
- Forecasting model (the centerpiece): turns planned mailing volume into the exact number of agents needed on any future day — a cohort-based simulation of call attempts, pickup decay, and callback flows. Because it runs off the mailing calendar, agent demand is known 500+ days ahead, far beyond the notice period scheduling requires. Includes capacity-choke detection and a batch-splitting optimizer.
- Performance analytics: a percentile-based hourly pacing metric that makes agents comparable across campaigns with very different call lengths and success rates, combined with outcomes into a single performance index powering real-time TV rankings on the floor.

Student Researcher — Philips

Sep 2022 – Dec 2022 · Eindhoven, Netherlands

- Researched OneBlade market penetration among students; built linear models estimating adoption of a blade-subscription service across student segments.

Data Entry — Von der Heide

Jan 2019 – Aug 2019 · Mexico City, Mexico

- Cleaned and migrated the company database to a new CRM system, safeguarding data integrity throughout the transfer.

Selected Projects

Market-Gap Analysis: Online-Booking Adoption in Mexico City

- End-to-end data pipeline measuring online-booking adoption across 10,605 businesses: 722 structured API

queries, deduplication, two independent booking detectors, and public-registry enrichment. Measured 6.9% adoption; delivered a 9,878-row qualified dataset (89% with direct contact details) as a workbook and dashboard.

Weather Prediction-Market Research

2026 · Python, probability calibration, market microstructure

- Tested whether prediction markets price official weather observations efficiently: implied-probability extraction from order books, no-arbitrage coherence checks, repricing-latency measurement. Documented an honest null result.

Citania — Booking & Payments Platform

2025–2026 · TypeScript, PostgreSQL, React Native

- Solo-built multi-tenant booking and payments SaaS: 100+ integration tests, database-level slot exclusivity, two-way Google Calendar sync. Live in production.

Football Player Value Modelling

2024 · Python, statistical modelling

- Predicted market values on a multi-year, 100+ competition dataset compiled myself. Best model: $R^2 = 0.921$, RMSE \$2.23M.

Education

MSc Data Science & Society (Business Track) — Tilburg University

2023 – 2024

BSc Economics — Tilburg University

2020 – 2023

Languages

English (native), Spanish (native), Dutch (A2, actively learning)